COMPLETE STUDY NOTES

Cloud Computing

Modules I · II · III — Architecture, Services, Virtualization & Beyond

Module I — Fundamentals | Module II — Services & Stack | Module III — Infrastructure

MODULE I

Introduction to Cloud Computing

⏱ 7 Hours

1. DEFINING CLOUD COMPUTING

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

> **NIST Definition:** "Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."

Instead of owning their own computing infrastructure or data centers, companies can rent access to anything from applications to storage from a cloud service provider. The key idea: *pay only for what you use*, eliminating huge capital expenditures.

2. ESSENTIAL CHARACTERISTICS (NIST)

On-Demand Self-Service

A consumer can unilaterally provision computing capabilities (server time, network storage) automatically without requiring human interaction from the service provider.

Broad Network Access

Capabilities available over the network via standard mechanisms promoting use by heterogeneous platforms (phones, tablets, laptops, workstations).

Resource Pooling

Provider's resources pooled to serve multiple consumers using a multi-tenant model. Different physical and virtual

resources assigned and reassigned dynamically.

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Rapid Elasticity

Capabilities can be elastically provisioned and released (sometimes automatically) to scale rapidly outward and inward commensurate with demand.

Measured Service

Cloud systems automatically control and optimize resource use by leveraging a metering capability. Resource usage monitored, controlled, and reported — providing transparency.

3. COMPONENTS OF CLOUD COMPUTING

Client

The user-facing device (PC, mobile, thin client) that interacts with the cloud via browsers or applications.

Data Center

Collection of servers housing the application to which users request via internet. The "back-end" of the cloud.

Distributed Servers

Servers spread geographically, enabling high availability, fault tolerance, and reduced latency.

Network

The internet backbone connecting clients to cloud data centers. Includes LAN, WAN, VPN, and CDN infrastructure.

Virtualization Layer

Hypervisors and VMs that abstract physical hardware into logical, shareable units of compute, storage, and network.

Management Software

Orchestration platforms (e.g., OpenStack, Kubernetes) that automate provisioning, monitoring, scaling, and billing.

4. DEPLOYMENT MODELS

Private Cloud

Cloud infrastructure operated solely for a single organization. Greater control and security, but higher cost. Example: A bank's internal cloud.

Public Cloud

Infrastructure owned and operated by a third-party provider (AWS, Azure, GCP) delivered over the internet. Most scalable and cost-effective.

Hybrid Cloud

Combines private and public clouds, allowing data and applications to be shared between them. Best of both worlds — flexibility + security.

Community Cloud

Shared among several organizations with common concerns (security, compliance, jurisdiction). Example: Government agencies sharing a GovCloud.

5. SERVICE MODELS (THE CLOUD STACK)

> Think of the cloud stack as a building: **IaaS** is the foundation (land + bricks), **PaaS** is the frame (structure), **SaaS** is the fully furnished room (ready to use).

IaaS

Infrastructure as a Service — Virtual machines, storage, networking. Users manage OS and above. Examples: AWS EC2, Google Compute Engine.

PaaS

Platform as a Service — Development platforms, databases, middleware. Developer focuses on code only. Examples: Heroku, Google App Engine.

SaaS

Software as a Service — Fully functional software delivered via browser. Nothing to install. Examples: Gmail, Dropbox, Salesforce.

6. APPLICATIONS OF CLOUD COMPUTING

- Email (Gmail, Outlook)
- File Storage (Dropbox, Drive)
- Streaming (Netflix, YouTube)
- AI & ML Platforms
- Healthcare (EHR Systems)
- Banking & FinTech
- Online Gaming
- Mobile App Backends
- Scientific Research
- E-Commerce Platforms

7. BENEFITS OF CLOUD COMPUTING

1. **Cost Savings:** Eliminates capital expense of buying hardware/software. Pay-as-you-go pricing converts CapEx to OpEx.
2. **Speed & Agility:** Resources provisioned in minutes, not weeks. Enables rapid experimentation and innovation.
3. **Scalability:** Scale up/down on-demand. Handle traffic spikes (like Black Friday) without over-provisioning permanently.
4. **Global Reach:** Deploy applications worldwide in seconds using provider's global network of data centers.
5. **Productivity:** IT teams freed from patching hardware, racking servers. Focus shifts to more valuable business goals.

6. **Performance:** Cloud services run on worldwide networks of secure data centers upgraded to the latest generation of hardware.
7. **Reliability:** Data backup, disaster recovery, and business continuity are easier and less expensive.
8. **Security:** Cloud providers offer broad set of policies, technologies, and controls that strengthen security posture.

8. LIMITATIONS OF CLOUD COMPUTING

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Security & Privacy

Data stored externally — risk of breaches. Regulatory compliance (GDPR, HIPAA) may restrict where data can reside.

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Internet Dependency

Cloud requires reliable, fast internet. Poor connectivity = degraded performance or complete service unavailability.

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Vendor Lock-in

Migrating between cloud providers is complex. Proprietary services make it difficult to switch without major effort.

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Limited Control

Users cannot control underlying infrastructure. Updates pushed by the provider may break existing applications.

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Downtime Risk

Even top providers face outages. AWS, Azure, GCP have all had significant outages impacting millions of users.

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Unexpected Costs

Pay-as-you-go can spiral. Egress fees, storage fees, and idle resources can lead to "bill shock."

9. GRID COMPUTING

Grid computing is the use of widely distributed computer resources to reach a common goal. A computational grid can be thought of as a distributed system with non-interactive workloads that involve many files.

> **Key Idea:** Grid computing connects many individual computers across different organizations/locations

> into a single large virtual supercomputer — used for solving computationally intensive tasks.

Examples: SETI@home (searching for alien signals), CERN's LHC Computing Grid, BOINC projects, Folding@home (protein folding simulation).

Architecture: Resource brokers, schedulers, job queues. Jobs submitted to a meta-scheduler which dispatches them to available grid nodes.

10. GRID VS. CLOUD COMPUTING

ASPECT	GRID COMPUTING	CLOUD COMPUTING
Focus	Computation-intensive tasks	On-demand resource delivery
Resource Sharing	Across organizations (federated)	Within a provider (centralized)
Billing Model	Usually free / institutional	Pay-as-you-go (commercial)
Management	Decentralized	Centralized (provider-managed)
Homogeneity	Heterogeneous nodes	Often standardized hardware
Standards	OGSA, WSRF (open standards)	Proprietary APIs (mostly)
Scalability	Limited by federation	Near-infinite, elastic
Users	Scientists, researchers	Enterprises, developers, individuals
Example	CERN LHC Grid	AWS, Azure, Google Cloud

MODULE II

Cloud Architecture, Services & Applications

⏱ 7 Hours

1. EXPLORING THE CLOUD COMPUTING STACK

The cloud computing stack is a layered architecture where each layer builds upon the one below it, providing services to the layer above. Think of it as a vertical pipeline from raw hardware all the way up to end-user applications.

Composability

The ability to combine cloud services (APIs, microservices) like building blocks to create complex applications. Services remain loosely coupled, independently deployable, and reusable.

Infrastructure

Physical layer: servers, storage, networking. Virtualized to become logical, on-demand resources. Foundation of every cloud service.

Platforms

Middleware, runtimes, databases, and developer tools. Provides the environment to build, run, and manage

applications without worrying about infrastructure.

Virtual Appliances
Pre-configured virtual machine images bundling an OS, application, and its dependencies — ready to deploy instantly. Examples: a pre-configured WordPress VM, a database appliance.

Communication Protocols
Standards enabling components to talk: HTTP/HTTPS (REST APIs), SOAP (legacy web services), AMQP (message queuing), WebSockets (real-time), gRPC (microservices).

Applications
Top layer — the software users directly interact with. SaaS applications like Gmail, Zoom, Salesforce live here. Entire stack below powers them invisibly.

2. INFRASTRUCTURE AS A SERVICE (IAAS)

IaaS provides virtualized computing resources over the internet. It is the most flexible cloud service model, giving users maximum control over the infrastructure.

> **What you get:** Virtual machines, storage (block, object), virtual networks, load balancers, firewalls.
> **What you manage:** OS, runtime, middleware, data, applications.
> **Provider manages:** Physical servers, network, data center, hypervisor.

AWS EC2 | Azure Virtual Machines | Google Compute Engine | DigitalOcean Droplets | IBM Cloud VMs

Use cases: Hosting websites, running legacy applications, big data analysis, backup and recovery, dev/test environments.

Advantages: Maximum flexibility, pay-per-use, no hardware procurement, disaster recovery built-in.

Disadvantages: User responsible for OS patching and security, requires technical expertise.

3. SOFTWARE AS A SERVICE (SAAS)

SaaS delivers a complete software application over the internet on a subscription basis. The cloud provider manages everything — hardware, OS, middleware, runtime, data, and the application itself.

> **What you get:** Fully functional application accessible via browser/app.
> **What you manage:** Your data, user settings, and access control.
> **Provider manages:** Everything else — all layers of the stack.

Gmail / Outlook 365 | Salesforce CRM | Zoom / Teams | Dropbox / Google Drive | Slack | Shopify

Advantages: Zero installation, accessible from anywhere, automatic updates, lower upfront costs, multi-device support.

Disadvantages: Limited customization, data privacy concerns, vendor lock-in, internet dependency.

4. PLATFORM AS A SERVICE (PAAS)

PaaS provides hardware and software tools over the internet. It gives developers everything needed to build cloud-based apps without managing infrastructure. Developers write code; the platform handles

deployment, scaling, load balancing, etc.

> **What you get:** Runtime environments, development frameworks, databases, CI/CD tools, APIs.
> **What you manage:** Application code and data only.
> **Provider manages:** OS, middleware, runtime, servers, storage, networking.

Heroku | Google App Engine | Azure App Service | AWS Elastic Beanstalk | Red Hat OpenShift

Use cases: Rapid application development, API development, microservices, database management, collaboration platforms.

Advantages: Faster development, reduced complexity, automatic scaling, built-in DevOps tools.

Disadvantages: Less control than IaaS, potential vendor lock-in at the platform level, runtime limitations.

5. IDENTITY AS A SERVICE (IDAAS)

IDaaS is a cloud-based service that provides identity and access management (IAM) capabilities. It handles authentication, authorization, and user management as a managed service, eliminating the need to build and maintain your own identity infrastructure.

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Single Sign-On (SSO)

One set of credentials to access multiple applications. Log in once, access everything — Gmail, Slack, Salesforce without re-authenticating.

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Multi-Factor Authentication

Additional verification steps (OTP, biometrics, push notifications) beyond password to verify identity.

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Directory Services

Central repository of user identities. Manages who is who in the organization — roles, groups, permissions.

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Access Governance

Policies controlling what resources each identity can access, when, and from where. Role-based access control (RBAC).

Okta | Microsoft Entra ID (Azure AD) | Auth0 | Google Workspace Identity

6. COMPLIANCE AS A SERVICE (CAAS)

Compliance as a Service refers to cloud-delivered tools and services that help organizations meet regulatory, legal, and industry-specific compliance requirements — automatically and continuously.

Why it matters: Regulations like GDPR, HIPAA, PCI-DSS, SOX require organizations to prove data handling practices meet strict standards. CaaS automates evidence collection, audit trails, and policy enforcement.

Audit Automation

Automatically collect evidence of compliance — logs, access records, configuration snapshots — and generate audit-ready reports.

Continuous Monitoring

Real-time checks against compliance frameworks. Alerts when systems drift from compliant configurations.

Policy Enforcement

Automatically block non-compliant actions (e.g., unencrypted data storage) before they occur.

Risk Assessment

Identify gaps in compliance posture and prioritize remediation efforts based on risk level.

Vanta | Drata | Tugboat Logic | AWS Audit Manager

7. SERVICE MODEL RESPONSIBILITY COMPARISON

LAYER	ON-PREMISES	IAAS	PAAS	SAAS
Applications	User	User	User	Provider
Data	User	User	User	Provider
Runtime	User	User	Provider	Provider
Middleware	User	User	Provider	Provider
OS	User	User	Provider	Provider
Virtualization	User	Provider	Provider	Provider
Servers	User	Provider	Provider	Provider
Storage	User	Provider	Provider	Provider
Networking	User	Provider	Provider	Provider

MODULE III

Cloud Infrastructure & Virtualization

⏱ 8 Hours

1. HARDWARE AND INFRASTRUCTURE OVERVIEW

Cloud infrastructure is the collection of hardware and software components — servers, storage, networking, virtualization software — that enables cloud computing. It can be thought of as the "physical

plant" hidden behind the cloud's elegant abstractions.

> A modern cloud data center contains tens of thousands of commodity servers organized into racks, rows, and pods — interconnected by massive switching fabrics capable of petabits per second of throughput.

2. CLIENTS

Cloud clients are the devices through which users access cloud services. The client is responsible for the user interface; all processing happens in the cloud data center.

Thick Client

Traditional desktop/laptop with full local computing power. Runs native apps that connect to cloud for data/sync. Examples: desktop Excel with OneDrive.

Thin Client

Minimal local processing — merely renders the UI. All computation done server-side. Cheaper, easier to manage. Used in VDI environments.

Mobile Client

Smartphones and tablets. Native apps or mobile browsers connect to cloud APIs. Unique challenges: intermittent connectivity, battery, screen size.

Browser Client

Web browser as the universal client. HTML5, JavaScript, WebAssembly enable rich apps without installation. Platform-independent, always up-to-date.

Trend: The shift toward browser and mobile clients has driven cloud adoption — any device, anywhere, anytime.

3. SECURITY IN CLOUD INFRASTRUCTURE

Cloud security is a shared responsibility model — the provider secures the cloud, the customer secures what's in the cloud. Understanding this boundary is critical.

Physical Security

Data centers protected by biometric access, CCTV, security guards, man-traps, and geographical redundancy. Users never need to worry about this layer.

Network Security

DDoS protection, firewalls, intrusion detection/prevention systems (IDS/IPS), encryption in transit (TLS/SSL). Virtual Private Clouds (VPCs) isolate tenant traffic.

Data Security

Encryption at rest (AES-256) and in transit. Key management services (KMS). Data residency controls for regulatory compliance.

Identity & Access

IAM policies, MFA, principle of least privilege, role-based access control (RBAC). Federation with enterprise directories.

Application Security

Web Application Firewalls (WAF), SAST/DAST scanning, container image scanning, secret management (no hardcoded credentials).

Compliance & Auditing

Continuous monitoring, audit logs (CloudTrail, etc.), SOC2/ISO27001 certifications of the provider, GDPR/HIPAA/PCI-DSS controls.

> **Shared Responsibility Model (AWS example):** AWS secures compute, storage, database, networking, and the global infrastructure. The *customer* secures OS configurations, network traffic, firewall rules, application code, identity management, and data encryption.

4. NETWORK IN CLOUD INFRASTRUCTURE

Cloud networking is the foundation enabling multi-tenant isolation, global connectivity, and high performance. It spans physical switching fabrics, software-defined networking, and edge delivery.

Virtual Private Cloud (VPC)

Logically isolated section of the cloud where you launch resources in a virtual network you define. Your own private IP address range, subnets, route tables.

Load Balancing

Distributes incoming traffic across multiple servers, preventing any single instance from becoming a bottleneck. Layer 4 (TCP) and Layer 7 (HTTP) balancers.

Content Delivery Network

Edge servers globally cached close to users, reducing latency for static assets. Examples: Cloudflare, AWS CloudFront, Akamai.

VPN & Direct Connect

Encrypted tunnels (VPN) or dedicated leased lines (AWS Direct Connect, Azure ExpressRoute) connecting on-premises infrastructure to the cloud securely.

Software-Defined Networking

SDN decouples network control from forwarding plane. Network programmable via APIs — enables rapid, automated network reconfiguration without physical changes.

DNS & IP Management

Cloud-native DNS services (Route 53, Azure DNS) with health checks, failover, geolocation routing. Elastic IPs that persist across instance stops/starts.

5. CLOUD SERVICES LAYER

Beyond raw compute and storage, cloud providers offer managed services that eliminate the operational burden of running common infrastructure components.

Managed Databases
RDS (MySQL, PostgreSQL), DynamoDB (NoSQL), CosmosDB. Automatic backups, patching, replication, and scaling — zero DBA work.

Serverless Computing
AWS Lambda, Azure Functions — run code without provisioning servers. Event-driven, millisecond billing, automatic scaling to zero.

Container Services
ECS, EKS (Kubernetes), Azure AKS — orchestrate containerized applications. Package once, run consistently anywhere.

AI/ML Services
SageMaker, Azure ML, Vertex AI — managed machine learning pipelines, pre-trained models, model deployment without infrastructure management.

Messaging Queues
SQS, SNS, Azure Service Bus — decouple components. Producer publishes message; consumer processes asynchronously. Enables fault-tolerant architectures.

6. VIRTUALIZATION TECHNOLOGY

Virtualization is the process of creating a software-based (virtual) representation of hardware resources. It is the fundamental enabling technology of cloud computing — without virtualization, there is no multi-tenancy, no elasticity, no on-demand provisioning.

> **Core Concept:** A *hypervisor* (Virtual Machine Monitor) runs on physical hardware and allows multiple operating systems to run simultaneously on a single physical machine, each isolated in its own Virtual Machine (VM).

TYPES OF HYPERVISORS

Type 1 — Bare Metal

Runs directly on hardware. No host OS layer. Maximum performance and security. Examples: VMware ESXi, Microsoft Hyper-V, Xen (used by AWS). *Used in all major cloud data centers.*

Type 2 — Hosted

Runs on top of a host OS. Slightly less efficient. Used for development and testing. Examples: VirtualBox, VMware Workstation, Parallels. *Not used in production cloud.*

TYPES OF VIRTUALIZATION

Server Virtualization

Multiple VMs on one physical server. Each VM has its own OS, CPU, memory allocation. The most fundamental form. Enables multi-tenancy.

Storage Virtualization

Pools physical storage from multiple devices into a single logical storage unit. Users see one large drive; the system manages physical placement transparently.

Network Virtualization

Combines hardware and software resources into a software-defined network. VLANs, SDN, NFV (Network Function Virtualization) — routers/firewalls as software.

Desktop Virtualization (VDI)

Virtual Desktop Infrastructure — user's desktop runs as a VM in the data center. Thin clients stream the display. Centralized management, any-device access.

Application Virtualization

Applications run in isolated environments, packaged with their dependencies. No installation conflicts. Examples: Docker containers, Citrix XenApp.

OS-Level Virtualization (Containers)

Multiple isolated user-space instances (containers) on a single OS kernel. Lighter than VMs — no separate OS per container. Docker, LXC, podman.

VM VS. CONTAINER

ASPECT	VIRTUAL MACHINE	CONTAINER
Isolation	Full OS isolation	Process-level isolation
Boot Time	Minutes	Seconds / milliseconds
Size	GBs (includes full OS)	MBs (shares host OS kernel)
Overhead	Higher (each OS uses resources)	Very low overhead
Security	Stronger isolation	Weaker (shared kernel)

Use Case	Different OS, strong isolation	Microservices, CI/CD pipelines
Examples	VMware, VirtualBox, EC2 instances	Docker, Kubernetes pods

BENEFITS OF VIRTUALIZATION IN CLOUD

1. **Resource Utilization:** Physical servers run at 10-15% without virtualization. With VMs, utilization jumps to 70-80%, drastically reducing hardware costs.
2. **Isolation:** Each tenant's VM is isolated. A crash in one VM doesn't affect others. Security boundaries enforced by the hypervisor.
3. **Rapid Provisioning:** New VMs deployed from templates in seconds. Previously took weeks (ordering hardware, racking, cabling, OS install).
4. **Live Migration:** VMs can be moved between physical hosts with zero downtime (VMware vMotion, Xen live migration). Enables maintenance without service interruption.
5. **Snapshots:** Capture VM state at a point in time. Roll back instantly if an update breaks something. Fundamental to cloud backup strategies.
6. **Elasticity:** Spin up 1,000 VMs to handle peak load; terminate them when load drops. Virtualization makes pay-per-use economically viable.

7. PUTTING IT TOGETHER — CLOUD ARCHITECTURE

A complete cloud architecture combines all three modules into a cohesive system:

Zone / Region
Physical data center location. Multiple Availability Zones per region for redundancy. Regions separated geographically for disaster recovery.

Compute Cluster
Racks of servers running hypervisors, managed by orchestration software. VMs and containers spawn on demand from a shared resource pool.

Storage Backend
Distributed storage systems (Ceph, S3, EBS). Block storage for VMs, object storage for files/media, file storage for shared network access.

Network Fabric
Spine-leaf switching topology, VXLAN overlays for VM isolation, BGP routing between PoPs, DDoS scrubbing at the edge.

Control Plane
Orchestration system (OpenStack, VMware vCenter, AWS control plane) — the brain. Accepts API calls, schedules resources, tracks state, enforces policies.

Management & Monitoring
Metrics, logs, traces collected from all layers. Dashboards (CloudWatch, Prometheus+Grafana). Auto-scaling triggers. Cost allocation and billing.

The cloud is just someone else's computer — understood deeply.

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